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Molecular detection and risk factors associated with multidrug-resistant *Campylobacter jejuni* from broiler cloacal and meat samples in Bangladesh

Md. Sirazul Islam¹  | Farazi Muhammad Yasir Hasib^{1,2} | Chandan Nath³ | Jahan Ara⁴ | Tahia Ahmed Logno³ | Md. Helal Uddin⁵ | Md. Ibrahim Khalil⁴ | Pronesh Dutta⁵ | Tridip Das⁶  | Sharmin Chowdhury^{1,4}

¹Department of Pathology and Parasitology, Faculty of Veterinary Medicine, Chattogram Veterinary and Animal Sciences University, Chattogram, Bangladesh

²Department of Infectious Diseases and Public Health, City University of Hong Kong, Hong Kong SAR, China

³Department of Microbiology and Veterinary Public Health, Faculty of Veterinary Medicine, Chattogram Veterinary and Animal Sciences University, Chattogram, Bangladesh

⁴One Health Institute, Chattogram Veterinary and Animal Sciences University, Chattogram, Bangladesh

⁵Department of Medicine and Surgery, Faculty of Veterinary Medicine, Chattogram Veterinary and Animal Sciences University, Chattogram, Bangladesh

⁶Poultry Research and Training Centre, Chattogram Veterinary and Animal Sciences University, Chattogram, Bangladesh

Correspondence

Sharmin Chowdhury, Department of Pathology and Parasitology, Faculty of Veterinary Medicine, Chattogram Veterinary and Animal Sciences University, Khulshi, Chattogram 4225, Bangladesh.

Email: sharminchowdhury@cvasu.ac.bd

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Abstract

The gastrointestinal tract of poultry is a potential source of *Campylobacter jejuni*. Here, the prevalence, risk factors, antimicrobial susceptibility profile and genetic relationship of *C. jejuni* were studied in broilers from farms and meat from live bird markets (LBMs) and super shops (SS). Pooled cloacal samples were obtained from farms in six districts of Bangladesh between June 2019 and March 2020. Pooled meat samples were obtained from LBMs and SS in the Chattogram district. Microbial culture, polymerase chain reaction (PCR), antimicrobial susceptibility tests were used to detect multidrug-resistant *C. jejuni*. A positive PCR amplicon was validated by *mapA* partial gene sequencing and subsequent phylogenetic analysis. In total, 12.5% (95% CI: 8.5–17.7%) of farms ($N = 216$) and 27.1% (95% CI: 15.28–41.85%) of LBMs and SS ($N = 48$) tested positive for *C. jejuni*. Moreover, 98% of the isolates were multidrug-resistant, with 86% resistant to five or more antimicrobial groups. Multivariable logistic regression analysis showed a downtime of <14 days, no separate footwear for shed access, and more than one person entering the sheds were significantly associated with *C. jejuni* colonization. Phylogenetic analysis revealed a strong relationship between *C. jejuni* strains obtained in Bangladesh and strains isolated in India, South Africa and Grenada from humans, pigs and bats. This study revealed significant contamination of broiler meat with *Campylobacter* spp. and *C. jejuni*. Potential sources of contamination and anthropogenic factors associated with the alarming prevalence of *C. jejuni* identified in this study would aid in reducing the growing risks of broiler-associated pathogens.

KEYWORDS

Campylobacter spp., multidrug resistance, prevalence, public health, zoonoses

1 | INTRODUCTION

Food-borne illnesses are a growing public health concern worldwide (Zeng et al., 2016). *Campylobacteriosis* is a bacterial food-borne disease associated with multiple sources, including poultry meat and products (Wieczorek et al., 2012). Of all *Campylobacter* species, *C. jejuni* is the most common species, accounting for 77.3% of campylobacteriosis cases (Doyle and Erickson, 2006). *C. jejuni* typically exists as a commensal in the intestinal tract of poultry (Williams et al., 2016), also have pathogenic properties that causes acute gastroenteritis and several clinical conditions in humans, such as reactive arthritis, pancreatitis, enterocolitis and bacteraemia, primarily owing to the consumption of undercooked poultry (Logue et al. 2017). *Campylobacter* spp. mainly colonize the cecum and colon of chickens, cecal contents can contaminate the meat and other parts during processing (Sakaridis et al., 2018). Moreover, the poultry farm environment is frequently contaminated, further amplifying the risk of transmission to humans. *Campylobacteriosis* is caused by poor sanitation and hygiene practices in the community, including children, in developing countries, including Bangladesh (Pollett et al., 2012; Taniuchi et al., 2013).

Campylobacteriosis is not the only public health issue associated with *Campylobacter* species; the emergence of antimicrobial resistance among *Campylobacter* species overrides the disease concern. Antimicrobial resistance, particularly multidrug resistance (MDR), has become a significant concern worldwide in pathogenic and environmental microorganisms. MDR is frequently found in food-borne pathogens. Neogi et al. (2020) reported that 9% of *C. jejuni* and 42% of *C. coli* were resistant to three or more antibiotics in poultry in Bangladesh. Misuse of antimicrobials may result in increasing selective pressure favouring antibiotic resistant strains but it is not the only cause and the relative importance is very difficult to determine. Misuse of antimicrobials in food-producing animals might be one of the important causes of resistance in food-borne pathogens, including *Escherichia coli*, *Salmonella*, *Enterococcus* and *Campylobacter* spp. (Akinbowale et al., 2006; Abraham et al., 2020). In recent decades, MDR *Campylobacter* strains have been identified in humans with gastroenteritis and diarrhoea in both developed and developing countries (Pollett et al., 2012; Mukherjee et al., 2014).

In Bangladesh, antimicrobials are frequently used in the poultry sector, therapeutically and prophylactically. However, the actual quantity of antimicrobials used is unknown, and there are no national sales statistics available (Imam et al., 2020). Moreover, most poultry farmers obtain antimicrobials directly from feed and drug sellers and pharmaceutical representatives without the involvement of registered veterinarians, indicating an absence of a regulatory regimen on antimicrobial use. Owing to the absence of specific guidelines and regulations, antimicrobial misuse has become a frequent phenomenon in poultry production in Bangladesh (Ferdous et al., 2019; Imam et al., 2020).

Introduction and spread of *Campylobacter* in broiler farms are believed to be predominantly associated with several farm management factors identified by prior studies in different countries,

Impacts

- Broiler colonization by zoonotic *Campylobacter jejuni* pose a considerable public health risk, as a significant proportion of these isolates are multidrug-resistant and may be transmitted to humans through the food chain.
- A striking 98% of the isolates were determined to be multidrug-resistant (resistant to at least three antimicrobial groups), with 86% showing resistance to five or more antimicrobial groups.
- This study identifies the risk factors from farm and live bird market management associated with biosecurity that need to be further investigated to understand and reduce the occurrence of multidrug-resistant *C. jejuni* in broiler flocks.

control programs are often challenged due to its multi-host predilection and survival in diverse environmental conditions though. Potential risk factors that might be considered in control programs, such as the presence of infected neighbouring farms within 2 km, increased flock age (>35 days) and size (>25,000 birds), more than one person entering the broiler house, use of hygienic measures (less frequent change of disinfectant in foot bath, absence of boot dips among others), summer season and positive colonization status of previous flocks, were suggested by previous studies (Barrios et al., 2006; McDowell et al., 2008; Chowdhury et al., 2012a; Chowdhury et al., 2012b; Sandberg et al., 2015; Høg et al., 2016). Moreover, farm management factors like less than 9 days of downtime between flocks, older broiler houses, disposal of litter within farm area, multiple broiler houses in the farm, mixed farming, staggered slaughter etc. (Hald et al., 2000; Guerin et al., 2007; Lyngstad et al., 2008; Chowdhury et al., 2012b) were ascertained as risk factors. Owing to dissimilar management and production systems, the risk factors for *Campylobacter* introduction and spread within and between farms may differ among countries, especially between developed and developing countries. Therefore, targeted and country-specific studies are of significant importance.

In Bangladesh, few studies have been conducted on *Campylobacter* colonization in broilers (Islam et al., 2018; Hasan et al., 2020). Furthermore, fewer studies have focused on discovering the antimicrobial susceptibility pattern of *Campylobacter* spp. in Bangladesh (Neogi et al., 2020). Molecular analyses of *Campylobacter* at the species level in poultry farms and live bird markets (LBMs) have not been thoroughly undertaken (Kabir, Islam, et al., 2014; Kabir, Sumon, et al., 2014; Sattar et al., 2014). In low-income countries, hygiene measures are often compromised at most levels of the broiler production system. Therefore, reducing colonization at the primary level (farm) could significantly impact public health. Recently, due to Bangladesh's economic transition to a developing country, growing concern for food safety, in terms of presence of zoonotic and drug resistant pathogens in food, has been observed. Moreover, in the last decade broiler intake

increased to a significant level that influenced a linear growth in the number of broiler farms in the country. Therefore, public health concern on food-borne pathogens, the role of broiler in disseminating campylobacter infection to humans, the knowledge gaps and subsequent, the present study aimed to assess the prevalence, risk factors, antimicrobial susceptibility profile and molecular and phylogenetic characteristics of *C. jejuni* isolated from poultry farms, LBMs and super shops in several districts of Bangladesh.

2 | METHODS

2.1 | Study design and sampling

The study used a cross-sectional design; between July 2019 and March 2020, cloacal swabs were collected from 216 commercial broiler chicken farms in six districts of Bangladesh (Chattogram, Dhaka, Khagrachari, Narsingdi, Narayanganj and Munshigonj), and 48 pooled meat samples were collected from LBMs and super shops (SS) in the Chattogram district. The sampling frame for the random selection of broiler farms was obtained from the government livestock office of each district. As all districts did not have an equal number of farms, proportionate stratified random sampling was done. To obtain a sample size of 200 farms, a proportion (20% to 25%) of all registered farms within a district were randomly selected. The sample size was calculated based on 15% prevalence of *C. jejuni* (Alam et al., 2020), 95% confidence limit, 5% margin of error and design effect 1. At each farm, cloacal swabs collected from five randomly sampled birds were subsequently pooled and placed in a Falcon tube containing buffered peptone water (BPW) (Oxoid™ Ltd., Basingstoke, UK). For meat samples, 8 SS in the Chattogram Metropolitan Area (CMA), 20 LBMs in the CMA and 20 LBMs from five peripheral subdistricts (Mirsorai, Fatikchari, Lohagara, Bashkhali and Rangunia) of Chattogram were randomly selected. At each LBM, one dressed broiler carcass per stall was collected from five randomly selected bird stalls. Similarly, one dressed broiler carcass was collected per SS. Thus, a total of 208 carcasses were collected. The carcasses were transported to the laboratory in an icebox. Five grams of samples (breast muscle, thigh muscle, liver and gizzard) were collected aseptically using sterile scissors and placed in separate Falcon tubes containing 45 mL BPW (Oxoid™ Ltd., Basingstoke, UK). All samples were kept at 4°C (max 24 h) and subsequently used for bacteriological investigation. Data on farm management, biosecurity and hygiene practices were collected through a face-to-face interview during sampling using a structured and pre-tested questionnaire (Appendix S1).

TABLE 1 Oligonucleotide primers used for identification of *Campylobacter* spp. and *C. jejuni*

Target gene	Primer sequence (5'-3')	Annealing temp (°C)	Product size (bp)	Reference
16S rRNA	F: ATCTAATGGCTTAACCATTAAC R: GGACGGTAACTAGTTTAGTATT	59	857	Denis et al. 1999
mapA	F: CTATTTTATTTTGAGTGCTTG R: GCTTTATTTGCCATTTGTTTTATTA	59	589	Denis et al. 1999

2.2 | Bacterial isolation

Campylobacter jejuni was isolated and identified from cloacal swabs and retail meat samples using standard bacteriological methods followed by molecular techniques. All BPW containing samples were incubated at 37°C for 16 hr followed by streaking on *Campylobacter* agar base plate (Oxoid™ Ltd.) containing selective growth supplement (Oxoid™ Ltd.) and 10% (v/v) defibrinated sheep blood (Vanderzant and Splittstoesser, 1992). The plates were incubated in an anaerobic jar (Oxoid™ AnaeroGen™ 2.5 L) under microaerophilic conditions with a CO₂ sachet (Thermo Scientific™ Oxoid AnaeroGen 2.5 L sachet; 10% CO₂, 95% humidity) at 42°C for 72 h.

2.3 | Phenotypic identification of *Campylobacter* isolates

A single isolated colony (small, round and creamy-grey) from each plate was microscopically examined for the characteristic appearance of *C. jejuni* using Gram staining (Boyer et al., 2021) and biochemically characterized using catalase and oxidase tests. The isolates were then kept in brain heart infusion broth (Oxoid™ Ltd.) containing 20% glycerol at -80°C until further use for molecular identification.

2.4 | Molecular identification of *Campylobacter* isolates

Genomic DNA was extracted using the conventional boiling-lysis method (Zhu et al., 2006). Molecular identification was performed by conventional PCR using the genus-specific primer for 16S rRNA gene and the *mapA* gene to detect *C. jejuni*; the primers are listed in Table 1. All PCR reactions were carried out on 15 µl of a final volume containing 2 µl (average of 3.45 ng/µl) DNA template, forward and reverse primer 1 µl each (20 picomoles/µl), 7.5 µl master mix (Thermo Fisher Scientific) and 3.5 µl nuclease-free water. The amplification conditions were as follows: initial denaturation at 95°C for 10 min, 35 cycles each consisting of 30 s at 95°C, 1 min 30 s at 59°C, 1 min at 72°C, and a final extension step of 10 min at 72°C (Denis et al., 1999). The PCR products were visualized on a Gel documentation system (UVP UVsolo touch - Analytik Jena AG) after electrophoresis with 1.5% agarose gel (SeaKem® LE Agarose from Lonza). *Campylobacter jejuni* subsp. *jejuni* ATCC 33560 strain and nuclease-free water were used as the positive and negative control.

2.5 | Determination of antimicrobial resistance of *Campylobacter jejuni*

Antimicrobial resistance was assessed using the disk diffusion method on Mueller-Hinton Agar (Oxoid™ Ltd.) enriched with 10% (v/v) defibrinated sheep blood. The incubation temperature was 42°C for 48 hr in a microaerophilic environment. *Campylobacter jejuni* isolates were tested against 12 antimicrobials belonging to seven groups: Group 1 - fluoroquinolones: ciprofloxacin (5 µg); Group 2 - cephalosporins: ceftriaxone (30 µg), cefotaxime (30 µg) and ceftazidime (30 µg); Group 3 - aminoglycosides: gentamicin (10 µg) and streptomycin (10 µg); Group 4 - tetracyclines: tetracycline (30 µg) and doxycycline (30 µg); Group 5 - sulphur drugs: sulfamethoxazole-trimethoprim (25 µg); Group 6 - macrolides: erythromycin (15 µg) and azithromycin (15 µg); Group 7 - penicillin: ampicillin (10 µg; Oxoid™ Ltd.). The isolates were categorized as susceptible, intermediate and resistant, based on the Clinical and Laboratory Standards Institute interpretative criteria (CLSI, 2020). Isolates resistant to at least three antimicrobial groups were considered multidrug-resistant (MDR) (Magiorakos et al., 2012). ATCC 25922 was used as a quality control organism.

2.6 | Nucleotide sequencing and phylogenetic analysis

The *mapA* gene amplicons of randomly selected six positive samples (from cloacal and meat) were purified using a DNA purification kit (Favorgen Biotech Corp, Taiwan) and sequenced by Sanger dideoxy sequencing using a commercial service (Macrogen Inc., Korea®). The sequence read data were manually cleaned up using the chromatograms and deposited in GenBank (accession number: MT175597–MT175599, MT920396–MT920398). Each sequencing read was subjected to NCBI BLASTn and aligned in Mega X using ClustalW (Kumar et al., 2018). A phylogenetic tree was constructed by the neighbour-joining method with a bootstrap value of 1000 replicates in Mega X. This analysis involved 25 nucleotide sequences where the outgroup of the phylogeny was *C. coli*. The accession numbers of the nucleotide sequences, retrieved from the Gene bank to compare the sequence data for this study are given in [Appendix S2](#).

2.7 | Ethics

Licensed veterinarians performed all assessments and sample collections from each sampling location. Before data collection and sampling, study participants, farmers and bird traders were informed about the aim of the research; participation was voluntary. The Chattogram Veterinary and Animal Sciences University's (CVASU) Ethical Committee (EC) reviewed the proposal and provided ethical approval on March 18, 2019 for conducting this research.

2.8 | Statistics

A farm/LBMs/SS was considered positive if a pooled sample was tested positive. In case of LBMs/SS any pooled sample either thigh/breast/liver/gizzard found positive is considered as a contaminated market. All data from the broiler farms, LBMs (urban and rural), and SS were imported into and coded in Excel (Microsoft Office 2016). Statistical analysis was performed using STATA-IC 13 software (StataCorp.). Cramer's V test, Spearman correlation coefficient, Chi-square test, t-test or ANOVA were used to evaluate the correlation and multicollinearity in categorical and numerical variables. Variables with a significant association or a Spearman correlation coefficient above 0.4 were considered correlated. Due to low sample size ($n = 9$) in Dhaka district (20% of the registered farms only), at data analysis stage it was merged with the nearest district, Narayanganj. Descriptive analysis was performed to estimate the prevalence and 95% confidence intervals (CI) of potential factors. Chi-square (χ^2) test was performed to determine the association of binary outcomes with hypothesized risk factors at the farm level, and a Fisher exact test was performed at the market level. Univariable and multivariable analyses were performed using logistic regression models to obtain the adjusted odds ratio of association between outcome and exposure variables. Factors in univariable analysis with a p -value of ≤ 1 were included in the multivariable model. The model was constructed using backward selection, and the accuracy of the fitted model was further evaluated using the receiver operating characteristics curve (ROC). Interactions between all explanatory variables (2 ways) were evaluated in the final model. A p -value of ≤ 0.05 in the final model was considered to indicate the potential risk factors of *C. jejuni* colonization. Maps displaying the geographical distribution of the farms, LBMs and SS with *C. jejuni* colonization were created using ArcGIS 10.8.

3 | RESULTS

3.1 | Prevalence of *Campylobacter jejuni*

In total, 264 pooled samples (cloacal swab and meat) were collected from the Chattogram, Dhaka, Narsingdi, Narayanganj, Munshigonj and Khagrachari districts of Bangladesh for *C. jejuni* isolation and identification. The geographical location of sampled broiler farms, LBMs and SSs, along with their colonization status, is illustrated in [Figure 1](#). By visual inspection, selected farms, LBMs and SSs locations and colonization status appear randomly distributed spatially. It was noted that, in some districts (e.g. Khagrachari, Narsingdi etc.) selected farm locations seem clustered as the baseline farms were majorly situated near transportation facilities, market locations and plain land. Fifty-six of the 216 farms (25.93%; 95% CI: 20.2–32.3%) were confirmed by PCR to be colonized with *Campylobacter*. 12.5% *C. jejuni* was isolated from cloacal samples (27/216; 95% CI: 8.5–17.7%; [Table 2](#)). Farm prevalence of *Campylobacter* spp. and *C. jejuni* by district is shown in [Figure 2](#). Additionally, the isolation percentage of *Campylobacter* in market was 54.17% (26/48; 95% CI: 39.17–68.63%); *Campylobacter jejuni* was found in 27.1% (13/48; 95% CI: 15.28–41.85%) of broiler

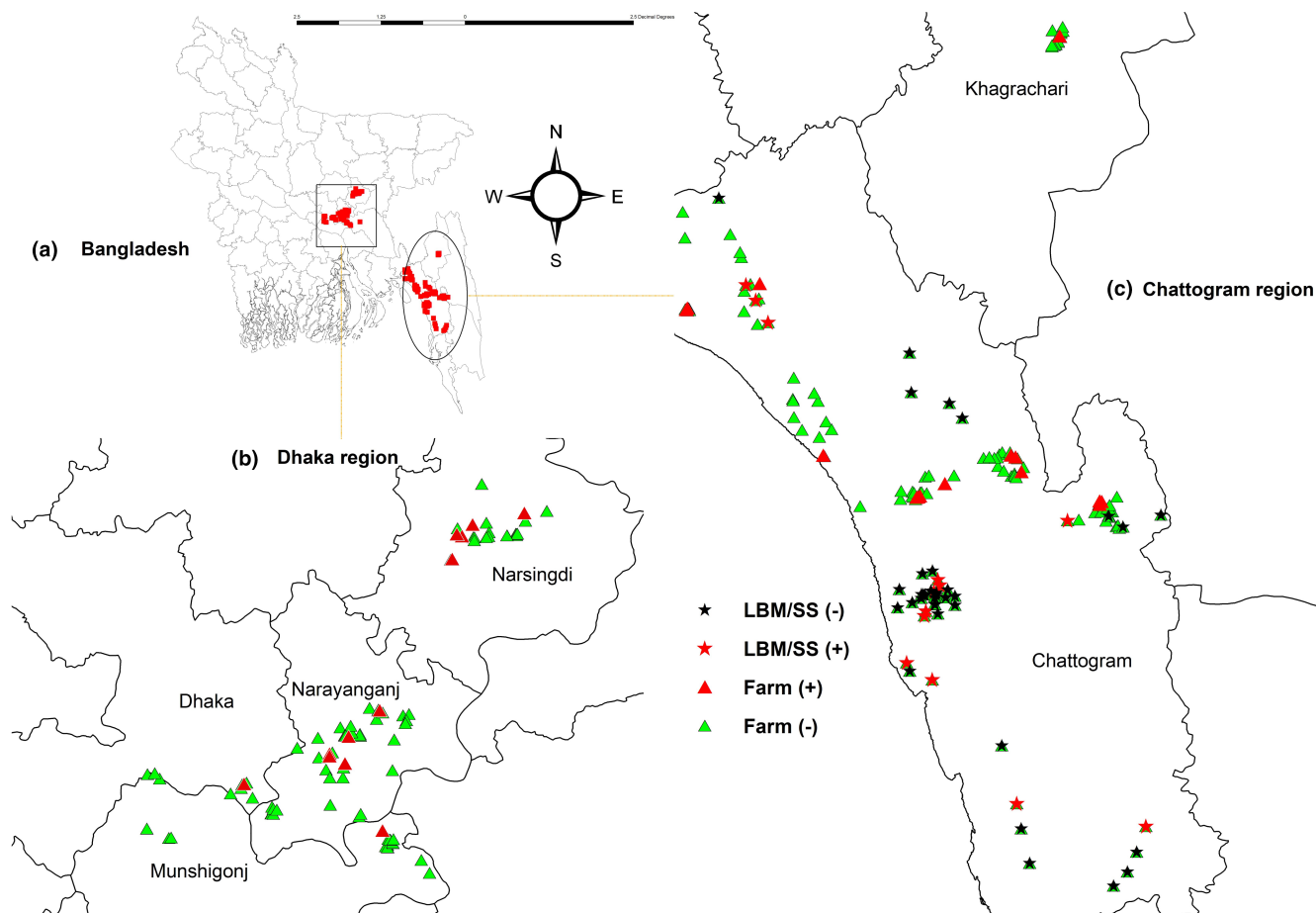


FIGURE 1 Map of Bangladesh displaying (a) the geographical location of poultry farms and markets included in the study, (b) the location of *Campylobacter jejuni*-positive and *C. jejuni*-negative farms in the Dhaka region and (c) *C. jejuni*-positive and *C. jejuni*-negative farms and markets in the Chattogram region

meat samples. Although 13 positive markets were identified as per our strategy, we obtained 15 isolates from the LBMs/SS. Altogether, 42 isolates (27 from 27 positive farms and 15 from 13 positive markets) were obtained from the collected sample.

3.2 | Antimicrobial resistance profile of *Campylobacter jejuni*

Antimicrobial susceptibility pattern of 42 *C. jejuni* isolates showed that all the isolates were resistant to streptomycin and erythromycin (100%) followed by azithromycin (97.62%; 95% CI: 87.43–99.99%), ciprofloxacin (85.71%; 95% CI: 71.46–94.57%), sulfamethoxazole-trimethoprim (80.95%; 95% CI: 65.88–91.4%), ceftazidime (71.43%; 95% CI: 55.42–84.28%), ceftriaxone (69.05%; 95% CI: 52.91–82.38%), cefotaxime (69.05%; 95% CI: 52.91–82.38%), tetracycline (59.52%; 95% CI: 43.28–74.37%) and ampicillin (57.14%; 95% CI: 40.96–72.27%). The highest number of the isolates showed sensitivity to doxycycline (69.05%; 95% CI: 52.91–82.38%) and gentamicin (59.52%; 95% CI: 43.28–74.37%). Alarming, 98% ($n = 41$) *C. jejuni* isolates were found to be MDR, 86% ($n = 36$) of them showed resistance against five or more antimicrobial groups (Figure 3).

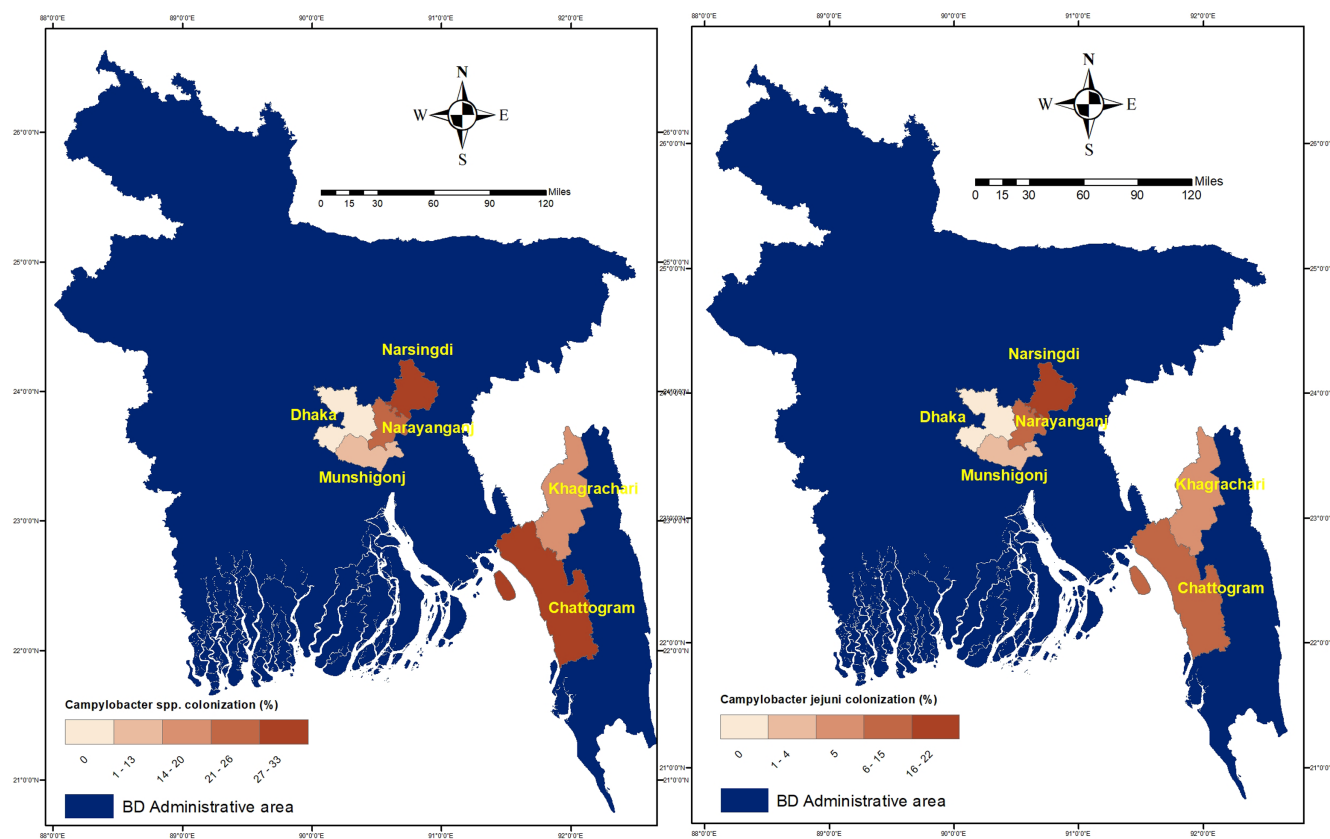
3.3 | Risk factor analysis

3.3.1 | Univariable association of risk factors with the occurrence of *Campylobacter* spp. and *Campylobacter jejuni* in broilers at farm level

The prevalence of *Campylobacter* spp. and *C. jejuni* associated with different farm-level variables in broilers are shown in Table 3. Several factors were identified as having a significant effect ($p \leq .1$) on the prevalence of *C. jejuni*. The prevalence of *C. jejuni* was higher in poultry sheds established before 2010 than in those established later (22.2% vs. 7.6%; OR = 3.4). The prevalence of *C. jejuni* was higher at farms where more than one person had access to the broiler shed than at farms where only one person had access to the farms (29.6% vs. 8.1%; OR = 4.8). The prevalence of *C. jejuni* was higher at farms that reared nine–ten flocks per shed per year than at farms that reared five–eight flocks per shed (14.5% vs. 4.6%; OR = 3.6). The prevalence was higher at farms with no separate footwear than at those with separate footwear (20.4% vs. 7%; OR = 3.4). Poultry farms that allowed downtime of ≤ 14 days between two flocks showed a higher prevalence of *C. jejuni* colonization than farms that allowed downtime of > 14 days (29.4% vs. 7.3%; OR = 5.3). Slaughter age of

TABLE 2 Prevalence of *Campylobacter* spp. and *C. jejuni* in broiler samples collected from diverse sources

Sample Source	Types of samples	<i>Campylobacter</i> spp.			<i>C. jejuni</i>		
		n	%	95% CI	n	%	95% CI
Farm (N = 216)	Cloacal swab	56	25.93	20.2–32.3	27	12.5	8.5–17.7
Live bird markets (N = 40)	Raw meat	25	62.5	45.8–77.27	12	30	16.56–46.53
Super shops (N = 8)	Raw meat	1	12.5	0.32–52.65	1	12.5	0.32–52.65

FIGURE 2 Map showing the status of *Campylobacter* spp. and *C. jejuni* farm prevalence in six districts of Bangladesh

>35 days was associated with a higher prevalence of *C. jejuni* than the slaughter age of ≤35 days (32.3% vs. 9.2%; OR = 4.7; Table 3).

3.3.2 | Multivariable logistic regression analysis to determine the potential risk factors associated with *Campylobacter* spp. and *Campylobacter jejuni* colonization status in broilers at farm level

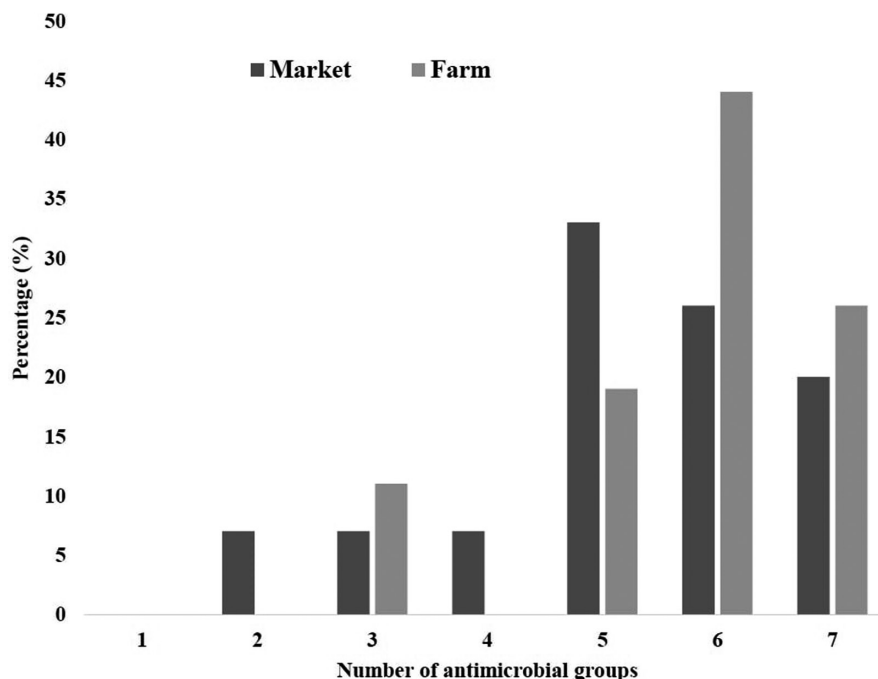
In the early models, area (farms within six districts) was added as a random effect. However, owing to low estimates, it was removed from further analysis. The multivariable logistic regression model identified three risk factors for *C. jejuni* colonization in broiler farms. The establishment year of broiler houses was identified as an additional risk factor (OR = 6.1, $p < .001$) in the final model for *Campylobacter* spp. colonization (Table 4). The odds of *C. jejuni*

colonization was 4.36 times higher at farms where sheds were kept empty for less than 14 days between two flocks ($p = .001$). Similarly, odds due to the entry of more than one person into the shed and not using separate footwear in the shed were 4.12 and 2.58 times higher, respectively ($p = .002$ and $.04$, respectively; Table 5).

3.3.3 | Univariable association of binary response of *Campylobacter* spp. and *Campylobacter jejuni* with potential explanatory variables in different markets in Chattogram

Isolation percentage of *C. jejuni* in meat samples according to market location, season, hygiene score, shop-kept birds and water source is shown in Table 6. Both rural and urban LBMs showed a 30% isolation percentage (95% CI: 11.89–54.28%; $N = 20$), whereas SS showed a

FIGURE 3 Multidrug resistance patterns of *Campylobacter jejuni* isolated from poultry farms and live bird markets



12.5% (95% CI: 0.31–52.65; $N = 8$; $p = .756$). The fisher-exact test showed no statistically significant association ($p > .05$) between different factors and the presence of *C. jejuni* in broiler meat (Table 6).

3.4 | Phylogenetic analysis

The phylogenetic tree developed to visualize the evolutionary comparison among the strains are displayed in Figure 4. The analysis showed that the six *C. jejuni* strains from which *mapA* gene sequences were obtained were closely clustered within the same clade. The tree also showed a close evolutionary relationship among strains isolated from humans, pigs and bats. Geographically, the isolated strains from this study showed a close relationship with strains from India, South Africa and Grenada.

4 | DISCUSSION

Poultry intestines are a favourable environment for the colonization of *Campylobacter*, thus increasing the likelihood of human campylobacteriosis due to the consumption of its contaminated meat, which is of great concern to human health (Kaakoush et al., 2015). *Campylobacter* is an extremely important food borne, zoonotic pathogen that infects millions of people each year worldwide. Human beings can get infections in different ways, but studies showed that broilers are the most important source (Mirzaie et al., 2011). Not many studies on *C. jejuni* in poultry in Bangladesh have been reported. Therefore, new scientific information is expected to be added to the existing literature by this study.

This study investigated the prevalence of *C. jejuni* and its antimicrobial resistance (especially MDR) pattern and potential risk factors

in poultry production and supply chain. In the present study, the overall farm-level prevalence (25.93%) of *Campylobacter* spp. is lower than those reported previously, ranging from 40% to 43% (Hasan et al., 2020; Neogi et al., 2020). However, Kabir, Islam, et al. (2014) reported a 78% prevalence in broilers at the flock level. It should be noted that the present study examined pooled cloacal samples from each farm. We hypothesize that screening cloacal samples at the individual level would have increased the proportion of *Campylobacter* isolation. *Campylobacter jejuni* was found in 12.5% of the samples in this investigation. Previous studies, conducted both nationally and internationally, validate the prevalence estimated in this study. *Campylobacter jejuni* was found in 17.9% and 6.25% of broiler samples in Bangladesh (Alam et al., 2020) and India (Malik et al., 2014). In contrast, Neogi et al. (2020) reported a higher occurrence (65%) at the broiler farm level in Bangladesh; however, the non-random farm selection and small sample size in the study of Neogi et al. (2020) may have influenced the prevalence estimation. In addition, the varied effects of season, rearing system, farm management, biosecurity and hygiene may influence the prevalence in different study populations (Cardinale et al., 2004; Guerin et al., 2007; Lyngstad et al., 2008; Näther et al., 2009; Sommer et al., 2013). In addition, laboratory techniques, settings and technicians' expertise in restricting contamination are significant factors for discrete results (Rahimi and Ameri, 2011; Vinuesa-Burgos et al., 2017).

We isolated *Campylobacter* from 54.17% of meat samples from 40 LBMs and 8 SS. The reported market-level isolation percentage of 54.17% *Campylobacter* spp. in chicken meat is comparable to the 49% recovery rate reported by Neogi et al. (2020). The present study found that 27.1% of raw chicken meat was contaminated with *C. jejuni*. The reported percentage differed from 68% estimated by Neogi et al. (2020) and 70.96% by Kabir, Sumon, et al. (2014) in Bangladesh. Numerous studies have reported the prevalence of

TABLE 3 Univariable analysis of potential factors associated with the *Campylobacter* spp. and *C. jejuni* colonization status of broiler farms (including variables with $p < .1$ only, $N = 216$)

Factor	Categories	Total number (N)	<i>Campylobacter</i> spp. (n)	OR (95% CI)	p	Total number (N)	<i>C. jejuni</i> (n)	OR (95% CI)	p
Establishment year of the house	After 2010	144	17	11.8	Ref	144	11	7.6	Ref
	Before 2010	72	39	54.2	8.8 (4.4–17.5)	72	16	22.2	3.4 (1.6–8.0)
Person entry into a shed	1	172	32	18.7	Ref	172	14	8.1	Ref
	>1	44	24	54.6	5.2 (2.6–10.7)	44	13	29.6	4.8 (2–11)
Flock per year	5–8	44	8	18.2	Ref	44	2	4.6	Ref
	9–10	172	48	27.9	1.7 (0.7–4)	172	25	14.5	3.6 (0.8–15.7)
Separate footwear	Yes	128	14	10.9	Ref	128	9	7	Ref
	No	88	42	47.7	7.4 (3.7–14.9)	88	18	20.4	3.4 (1.4–8)
Slaughter age (day)	≤35	185	39	21.1	Ref	185	17	9.2	Ref
	>35	31	17	54.8	4.5 (2.1–10)	31	10	32.3	4.7 (1.9–11.6)
Infected neighbour broiler farms	Present	27	3	11.1	Ref	27	3	11.1	Ref
	Absent	189	53	28	3.1 (1–10.8)	189	24	12.7	1.2 (0.3–4.1)
All in all-out system	Yes	166	37	22.3	Ref	166	18	10.8	Ref
	No	50	19	38	2.1 (1.1–4.2)	50	9	18	1.8 (0.7–4.3)
Length of downtime (days)	>14	165	26	15.8	Ref	165	12	7.3	Ref
	≤14	51	30	58.8	7.6 (3.8–15.3)	51	15	29.4	5.3 (2.2–12.3)

TABLE 4 Potential risk factors associated with *Campylobacter* spp. colonization in broiler farms (N = 216) identified using multivariable logistic regression (area under ROC curve = 0.8894)

Factor	Categories	OR	95% CI	p
Establishment year of the house	After 2010	Ref	2.66–13.97	<.0001
	Before 2010	6.10		
Person entry into a shed	1	Ref	1.51–9.15	.004
	>1	3.72		
Separate footwear	Yes	Ref	2.76–15.48	<.0001
	No	6.5		
Length of downtime	>14 days	Ref	2.75–15.58	<.0001
	≤14 days	6.54		

TABLE 5 Potential risk factors associated with *Campylobacter jejuni* colonization in broiler farms (N = 216) identified using multivariable logistic regression (area under ROC curve = 0.7964)

Factor	Categories	OR	95% CI	p
Person entry into a shed	1	Ref	1.68–10.13	.002
	>1	4.12		
Separate footwear	Yes	Ref	1.04–6.43	.04
	No	2.58		
Length of downtime	>14 days	Ref	1.80–10.57	.001
	≤14 days	4.36		

TABLE 6 Univariable association between different selected factors with the *Campylobacter* spp. and *Campylobacter jejuni* status of broiler meat samples collected from various sources in Chattogram district (fisher-exact test)

Variable	Category (n)	<i>Campylobacter</i> spp. %, (95% CI)	p (fisher-exact test)	<i>C. jejuni</i> %, (95% CI)	p (fisher-exact test)
Market location	Super shop (8)	12.5 (0.31–52.65)	.024	12.5 (0.31–52.65)	.756
	Rural (20)	55 (31.53–76.94)		30 (11.89–54.28)	
	Metro (20)	70 (45.72–88.11)		30 (11.89–54.28)	
Season	Spring (28)	42.86 (40.99–86.66)	.173	25 (10.69–44.87)	.586
	Winter (17)	70.59 (44.04–89.69)		35.29 (14.21–61.67)	
	Autumn (3)	66.67 (9.43–99.16)		0, (0–70.76)	
Hygiene score	Very clean (8)	12.5 (0.32–52.65)	.033	12.5 (0.31–52.65)	.742
	Moderate (27)	59.26 (38.8–77.61)		29.63 (13.75–50.18)	
	Dirty (13)	69.23 (38.57–90.91)		30.77 (9.1–61.42)	
Birds stay at the shop	One day (20)	50 (27.19–72.80)	.815	25 (8.66–49.1)	1
	Two days (17)	52.94 (27.81–77.02)		29.41 (10.31–55.96)	
	Three days (11)	63.64 (30.79–89.07)		27.27 (6.02–60.97)	
Water source	Tube well (14)	50 (23.04–76.96)	.842	35.71 (12.76–64.86)	.645
	Pond (6)	66.67 (22.28–95.67)		16.67 (0.42–64.12)	
	WASA (28)	53.57 (33.87–72.49)		25 (10.69–44.87)	

C. jejuni in poultry meat in different parts of the world, and the observed prevalence has varied. Ilida and Faridah (2012) reported a prevalence of 51.06% in chicken meat and chicken-based products in Malaysia. Skarp et al. (2016) reported an 11–71% prevalence of *Campylobacter* in retail poultry meat in European countries. A literature survey on *Campylobacter* in retail poultry in various countries reported a prevalence between 7.1% and 100% (Suzuki and Yamamoto, 2009). Several factors, including differences in colonization rates in food animals and production systems, cause these variations.

In this study, all the isolates showed high resistance against routinely used antimicrobials such as erythromycin, azithromycin, ciprofloxacin and tetracycline, which was higher than reported in previous studies from India (Kumar et al., 2021) and Bangladesh (Neogi et al., 2020). Moreover, it was reported that fluoroquinolones (e.g. ciprofloxacin) and macrolides (e.g. erythromycin, azithromycin) are the most common drugs used to treat *Campylobacter* infection in humans, and tetracycline is used as an alternative (Cantero et al., 2018). In contrast, resistance to aminoglycosides was variable in our study, with higher resistance to streptomycin (100%) and

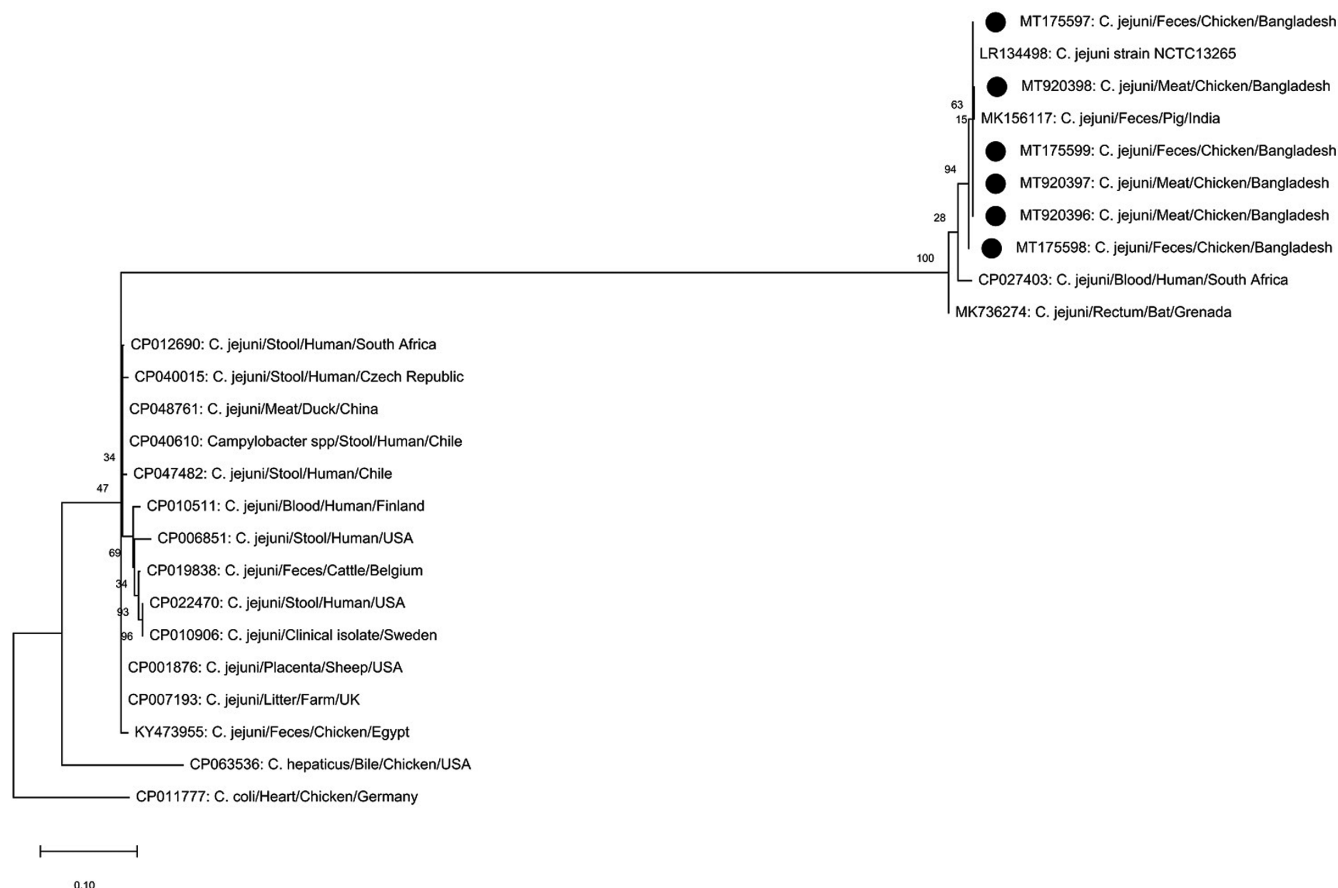


FIGURE 4 Phylogenetic analysis of the *mapA* gene of selected *Campylobacter jejuni* strains from this study and other global strains. The evolutionary history was inferred using the neighbour-joining method

lower resistance to gentamicin (40.5%), consistent with the findings of Cantero et al. (2018). Alarming, 100% of the isolates were found to be MDR, which is higher than previous findings in Bangladesh (49%; Neogi et al., 2020), India (53.2%; Kumar et al., 2021) and Pakistan (90.4%; Nisar et al., 2017). The high resistance pattern might be due to an absence of a regulatory regimen on antimicrobial use in Bangladesh.

The current study found that a house built before 2010 has a higher risk of *Campylobacter* colonization, related to other factors such as the kind of ventilation system and temperature regulating system (cooling and heating). Old ventilation systems may be ineffective at removing damp and moist air, allowing *Campylobacter* to thrive in the environment (Chowdhury et al., 2012b). Cleaning and disinfecting can be difficult in an old property with old damaged floors (McDowell et al., 2008).

In this study, we found entering more than one person into the broiler house as a potential risk factor related to *C. jejuni* colonization. As a result, it appears that environmental contamination throughout the production phase is the most likely source of *C. jejuni* colonization (Hald et al., 2000). The results of Chowdhury et al. (2012b) are consistent with this finding. Human contact is widely suspected to be a vital route (via boots, hands, clothes) for introducing *Campylobacter* into broiler houses (Cardinale et al., 2004;

Hald et al., 2000). Furthermore, molecular studies have also found evidence of subsequent colonization in broiler flocks with strains similar to those isolated from workers' shoes (Messens et al., 2009). The present study also identified a significant association between the use of separate footwear in the shed and the presence of *C. jejuni*. Previous research suggests that disinfecting shoes before entering the broiler sheds reduces the likelihood of *Campylobacter* colonization (Sibanda et al., 2018). However, the workers' hygiene practices should be considered because it may confound the association between *Campylobacter*'s introduction into the broiler house and the number of individuals who routinely enter the house.

The present study identified an interval of ≤ 14 days between two batches of broilers to be a risk factor. Theoretically, for a broiler shed to dry completely, extended downtime should be allowed so the farm environment can become least suitable for *Campylobacter* colonization. The finding is consistent with that reported by Lyngstad et al. (2008) and Hasan et al. (2020). In contrast, Høeg et al. (2016) indicated that an extended downtime implied a greater risk of colonization. Nonetheless, Barrios et al. (2006) observed that downtime did not affect the likelihood of *Campylobacter* colonization. *Campylobacter* spp., being chemotactic and aerotactic bacteria, can survive on the floor until it is properly dry and for an even longer time in the environment (Hald et al., 2000). Furthermore, Hazeleger

et al. (1998) reported that *Campylobacter* could survive at temperatures ranging from 4°C to 33°C. Oberheim et al., 2020 recently reported that *Campylobacter jejuni* could survive at low temperatures for up to 4 months.

The higher prevalence of *C. jejuni* in LBM meat samples than in broiler farms observed in this study is most likely owing to the floor environment, allowing for widespread secondary transmission of this zoonotic pathogen (Neogi et al., 2020). In Bangladesh, personal and environmental hygiene is rarely practiced in these LBMs. Even in developed countries with appropriate husbandry methods and health measures, *C. jejuni* contamination of poultry products in retail markets and slaughterhouses is a significant cause of food-borne illness (Kramer et al., 2000; Reich et al., 2018). We observed that the level of hygiene and biosafety practices—including regular hand washing, use of disinfectants, and cleaning of floors and cages—is lower in LBMs than in SS. The ideal slaughtering method can efficiently minimize the bacterial burden by limiting cross-contamination and washing chicken carcasses with chlorinated water (Pissol et al., 2013).

The analysis of the *mapA* gene sequencing data in this study revealed that all six sequenced strains belonged to the same clade, indicating that these strains have modest differences at the *mapA* gene level. Because the *mapA* gene was only partially sequenced, the phylogenetic tree cannot be used to assess their total clonal similarity or dissimilarity. Despite the various locations of sample collection observed in the constructed tree, there is a close evolutionary relationship among the strains. This may be because only the most common species were sampled. However, the strains' similarity to another strain from surrounding countries such as India suggests that these strains are well distributed regardless of their geographical location. Several prior instances of human infection by *C. jejuni* from poultry were predicted based on the phylogenetic tree, which showed that the *C. jejuni* strains of human origin were in the same cluster as the study strains (Friis et al., 2010). Overall, these partial gene sequencing results suggest evolutionary relatedness, but whole-genome sequencing will be required to verify these relationships.

Some limitations of the present study should be considered. We conducted the study only in a particular geographical region of Bangladesh, and a large part of the country was not included. A more detailed and comprehensive picture of the condition of the farms may be obtained if more farms from different regions of the country are included. This study investigated only broiler chickens; however, layer and backyard chickens are also a significant part of the poultry production system in Bangladesh. Finally, although *C. jejuni* was identified in the study, other *Campylobacter* spp. could not be studied.

5 | CONCLUSION

The study provides insights into epidemiology, that is, regarding the prevalence of *C. jejuni* at poultry farms and LBMs, along with the risk factors of *C. jejuni* colonization in commercial broiler chicken in Bangladesh. This study also revealed that broiler

samples were highly contaminated with *Campylobacter* spp. where a considerable number of strains were MDR *C. jejuni*. Phylogenetic analysis showed that *C. jejuni* isolated from various poultry sources are closely related. Similar to previous findings, our findings emphasize the importance of biosecurity measures in keeping chicken flocks free of pathogens. Based on the association of significant risk factors, motivational training programs are highly recommended for poultry handlers to maintain strict personal and environmental hygiene.

AUTHOR CONTRIBUTIONS

Md. Sirazul Islam involved in conceptualization, data curation, analysis, methodology, project investigation, NCBI data submission, visualization, writing—original draft, review and editing. **Farazi Muhammad Yasir Hasib** involved in data curation, methodology, writing—review and editing. **Chandan Nath** involved in data curation, methodology, formal analysis, writing—review and editing. **Jahan Ara** involved in data curation and methodology. **Tahia Ahmed Logno** involved in data curation and methodology. **Md. Helal Uddin** involved in data curation and formal analysis. **Md. Ibrahim Khalil** involved in data visualization. **Pronesh Dutta** involved in data visualization and formal analysis. **Tridip Das** involved in phylogenetic analysis. **Sharmin Chowdhury** involved in conceptualization, funding acquisition, project administration, supervision, validation, writing—review and editing.

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CONFLICT OF INTEREST

The authors have no potential conflict of interest.

DATA AVAILABILITY STATEMENT

The 16S rRNA gene of selected six *mapA* gene positive samples were initially sequenced for early molecular confirmation of *Campylobacter* spp. isolates. The sequence data has been submitted to the NCBI database under the accession numbers MT032361–MT032363 and MT912607–MT912609.

ORCID

Md. Sirazul Islam  <https://orcid.org/0000-0002-0525-8401>

Tridip Das  <https://orcid.org/0000-0001-5642-4150>

REFERENCES

- Abraham, S., Sahibzada, S., Hewson, K., Laird, T., Abraham, R., Pavic, A., Truswell, A., Lee, T., O'Dea, M., & Jordan, D. (2020). Emergence of fluoroquinolone-resistant *Campylobacter jejuni* and *Campylobacter coli* among Australian chickens in the absence of fluoroquinolone use. *Applied and Environmental Microbiology*, 86(8), e02765–e02719.

- Akinbowale, O. L., Peng, H., & Barton, M. D. (2006). Antimicrobial resistance in bacteria isolated from aquaculture sources in Australia. *Journal of Applied Microbiology*, 100(5), 1103–1113.
- Alam, B., Uddin, M., Mridha, D., Akhter, A. H. M., Islam, S. K., Haque, A. K. M., & Kabir, S. M. (2020). Occurrence of *Campylobacter* spp. in selected small scale commercial broiler farms of Bangladesh related to good farm practices. *Microorganisms*, 8(11), 1778.
- Barrios, P. R., Reiersen, J., Lowman, R., Bisaillon, J. R., Michel, P., Fridriksdóttir, V., Gunnarsson, E., Stern, N., Berke, O., McEwen, S., & Martin, W. (2006). Risk factors for campylobacter spp. colonization in broiler flocks in Iceland. *Preventive Veterinary Medicine*, 74(4), 264–278.
- Boyer, P. H., Talagrand-Reboul, É., Guffroy, A., Schramm, F., Jaulhac, B., & Grillon, A. (2021). Direct examination of gram-negative spiral-shaped rods: What is hidden behind the campylobacter spp.? *Clinical Microbiology and Infection*, 27(3), 396–397.
- Cardinale, E., Tall, F., Gueye, E. F., Cisse, M., & Salvat, G. (2004). Risk factors for campylobacter spp. infection in Senegalese broiler-chicken flocks. *Preventive Veterinary Medicine*, 64(1), 15–25.
- Cantero, G., Correa-Fiz, F., Ronco, T., Strube, M., Cerdà-Cuellar, M., & Pedersen, K. (2018). Characterization of *Campylobacter jejuni* and *Campylobacter coli* broiler isolates by whole-genome sequencing. *Foodborne Pathogens and Disease*, 15(3), 145–152.
- Chowdhury, S., Sandberg, M., Themudo, G. E., & Ersbøll, A. K. (2012a). The effect of presence of infected neighbouring farms for the campylobacter infection status in Danish broiler farms. *Spatial and Spatio-Temporal Epidemiology*, 3(4), 311–322.
- Chowdhury, S., Sandberg, M., Themudo, G. E., & Ersbøll, A. K. (2012b). Risk factors for campylobacter infection in Danish broiler chickens. *Poultry Science*, 91(10), 2701–2709.
- Clinical and Laboratory Standards Institute. (2020). *Performance standards for antimicrobial susceptibility testing. CLSI supplement M100*. Clinical and Laboratory Standards Institute.
- Denis, M., Soumet, C., Rivoal, K., Ermel, G., Blivet, D., Salvat, G., & Colin, P. (1999). Development of am-PCR assay for simultaneous identification of *Campylobacter jejuni* and *C. coli*. *Letters in Applied Microbiology*, 29(6), 406–410.
- Doyle, M. P., & Erickson, M. C. (2006). Reducing the carriage of food-borne pathogens in livestock and poultry. *Poultry Science*, 85(6), 960–973.
- Ferdous, J., Sachi, S., Al Noman, Z., Hussani, S. M., Sarker, Y. A., & Sikder, M. H. (2019). Assessing farmers' perspective on antibiotic usage and management practices in small-scale layer farms of Mymensingh district, Bangladesh. *Veterinary World*, 12(9), 1441–1447.
- Friis, C., Wassenaar, T. M., Javed, M. A., Snipen, L., Lagesen, K., Hallin, P. F., Newell, D. G., Toszeghy, M., Ridley, A., Manning, G., & Ussery, D. W. (2010). Genomic characterization of *Campylobacter jejuni* strain M1. *PLoS One*, 5(8), e12253.
- Guerin, M. T., Martin, W., Reiersen, J., Berke, O., McEwen, S. A., Bisaillon, J. R., & Lowman, R. (2007). A farm-level study of risk factors associated with the colonization of broiler flocks with campylobacter spp. in Iceland, 2001–2004. *Acta Veterinaria Scandinavica*, 49(1), 1–12.
- Hald, B., Wedderkopp, A., & Madsen, M. (2010). Thermophilic campylobacter spp. in Danish broiler production: A cross-sectional survey and a retrospective analysis of risk factors for occurrence in broiler flocks. *Avian Pathology*, 29(2), 123–131.
- Hasan, M. M., Talukder, S., Mandal, A. K., Tasmim, S. T., Parvin, M. S., Ali, M. Y., Sikder, M. H., & Islam, M. T. (2020). Prevalence and risk factors of campylobacter infection in broiler and cockerel flocks in Mymensingh and Gazipur districts of Bangladesh. *Preventive Veterinary Medicine*, 180, 105034.
- Hazeleger, W. C., Wouters, J. A., Rombouts, F. M., & Abee, T. (1998). Physiological activity of *Campylobacter jejuni* far below the minimal growth temperature. *Applied and Environmental Microbiology*, 64(10), 3917–3922.
- Høg, B. B., Sommer, H. M., Larsen, L. S., Sørensen, A. I. V., David, B., Hofshagen, M., & Rosenquist, H. (2016). Farm specific risk factors for campylobacter colonization in Danish and Norwegian broilers. *Preventive Veterinary Medicine*, 130, 137–145.
- Imam, T., Gibson, J. S., Foyals, M., Das, S. B., Gupta, S. D., Fournié, G., Hoque, M. A., & Henning, J. (2020). A cross-sectional study of antimicrobial usage on commercial broiler and layer chicken farms in Bangladesh. *Frontiers in Veterinary Science*, 7, 576113.
- Ilida, M. N., & Faridah, M. S. (2012). Prevalence of *Campylobacter jejuni* in chicken meat and chicken-based products. *Journal of Tropical Agriculture and Food Science*, 40(1), 63–69.
- Islam, M. K., Kabir, S. L., Haque, A. Z., Sarker, Y. A., & Sikder, M. H. (2018). Molecular detection and characterization of *Escherichia coli*, salmonella spp. and campylobacter spp. isolated from broiler meat in Jamalpur, Tangail, Netrokona and Kishoreganj districts of Bangladesh. *African Journal of Microbiology Research*, 12(32), 761–770.
- Kaakoush, N. O., Castaño-Rodríguez, N., Mitchell, H. M., & Man, S. M. (2015). Global epidemiology of campylobacter infection. *Clinical Microbiology Reviews*, 28(3), 687–720.
- Kabir, S. M. L., Islam, J., Suman, M. H., Khan, M. S. R., & Yamasaki, S. (2014). Isolation, identification and antimicrobial susceptibility profiles of campylobacter species with assessment of their risk factors in broiler flocks of Bangladesh agricultural university poultry farm. *Journal of Basic and Applied Scientific Research*, 4, 160–168.
- Kabir, S. M. L., Sumon, M. H., Amin, M. M., & Yamasaki, S. (2014). Isolation, identification and antimicrobial resistance patterns of *Campylobacter* species from broiler meat sold at KR Market of Bangladesh Agricultural University Campus, Mymensingh. *Journal of Agriculture and Food Technology*, 4(4), 15–21.
- Kramer, J. M., Frost, J. A., Bolton, F. J., & Wareing, D. R. (2000). *Campylobacter* contamination of raw meat and poultry at retail sale: Identification of multiple types and comparison with isolates from human infection. *Journal of Food Protection*, 63(12), 1654–1659.
- Kumar, S. M., Ramees, T. P., Dhanze, H., Gupta, S., Dubal, Z. B., & Kumar, A. (2021). Occurrence and antimicrobial resistance of campylobacter isolates from broiler chicken and slaughter house environment in India. *Animal Biotechnology*, 1–9. <https://doi.org/10.1080/10495398.2021.1953514>
- Kumar, S., Stecher, G., Li, M., Knyaz, C., & Tamura, K. (2018). MEGA X: Molecular evolutionary genetics analysis across computing platforms. *Molecular Biology and Evolution*, 35(6), 1547–1549.
- Logue, C. M., Barbieri, N. L., & Nielsen, D. W. (2017). Pathogens of food animals: Sources, characteristics, human risk, and methods of detection. *Advances in Food and Nutrition Research*, 82, 277–365.
- Lyngstad, T. M., Jonsson, M. E., Hofshagen, M., & Heier, B. T. (2008). Risk factors associated with the presence of campylobacter species in Norwegian broiler flocks. *Poultry Science*, 87(10), 1987–1994.
- Magiorakos, A. P., Srinivasan, A., Carey, R. B., Carmeli, Y., Falagas, M. E., Giske, C. G., Harbarth, S., Hindler, J. F., Kahlmeter, G., Olsson-Liljequist, B., Paterson, D. L., Rice, L. B., Stelling, J., Struelens, M. J., Vatopoulos, A., Weber, J. T., & Monnet, D. L. (2012). Multidrug-resistant, extensively drug-resistant and pandrug-resistant bacteria: An international expert proposal for interim standard definitions for acquired resistance. *Clinical Microbiology and Infection*, 18(3), 268–281.
- Malik, H., Kumar, A., Rajagunalan, S., Kataria, J. L., & Sachan, S. (2014). Prevalence of *Campylobacter jejuni* and *Campylobacter coli* among broilers in Bareilly region. *Veterinary World*, 7(10), 784–787.
- McDowell, S. W. J., Menzies, F. D., McBride, S. H., Oza, A. N., McKenna, J. P., Gordon, A. W., & Neill, S. D. (2008). *Campylobacter* spp. in conventional broiler flocks in Northern Ireland: Epidemiology and risk factors. *Preventive Veterinary Medicine*, 84(3–4), 261–276.
- Messens, W., Herman, L., De Zutter, L., & Heyndrickx, M. (2009). Multiple typing for the epidemiological study of contamination of broilers

- with thermotolerant campylobacter. *Veterinary Microbiology*, 138(1–2), 120–131.
- Mirzaie, S., Hassanzadeh, M., Bashashati, M., & Barrin, A. (2011). Campylobacter occurrence and antimicrobial resistance in samples from ceca of commercial turkeys and quails in Tehran, Iran. *International Research Journal of Microbiology*, 2(9), 338–342.
- Mukherjee, P., Ramamurthy, T., Mitra, U., & Mukhopadhyay, A. K. (2014). Emergence of high-level azithromycin resistance in *Campylobacter jejuni* isolates from pediatric diarrhea patients in Kolkata, India. *Antimicrobial Agents and Chemotherapy*, 58(7), 4248.
- Näther, G., Alter, T., Martin, A., & Ellerbroek, L. (2009). Analysis of risk factors for campylobacter species infection in broiler flocks. *Poultry Science*, 88(6), 1299–1305.
- Neogi, S. B., Islam, M. M., Islam, S. S., Akhter, A. T., Sikder, M. M. H., Yamasaki, S., & Kabir, S. L. (2020). Risk of multi-drug resistant campylobacter spp. and residual antimicrobials at poultry farms and live bird markets in Bangladesh. *BMC Infectious Diseases*, 20(1), 1–14.
- Nisar, M., Mushtaq, M. H., Shehzad, W., Hussain, A., Muhammad, J., Nagaraja, K. V., & Goyal, S. M. (2017). Prevalence and antimicrobial resistance patterns of campylobacter spp. isolated from retail meat in Lahore, Pakistan. *Food Control*, 80, 327–332.
- Oberheim, J., Höser, C., Lüchters, G., & Kistemann, T. (2020). Small-scaled association between ambient temperature and campylobacteriosis incidence in Germany. *Scientific Reports*, 10(1), 1–2.
- Pissol, A. D., De Oliveira, D., Toniazzi, G., Valduga, E., & Cansian, R. L. (2013). The effect of water pressure and chlorine concentration on microbiological characteristics of spray washed broiler carcasses. *Poultry Science Journal*, 1(2), 63–77.
- Pollett, S., Rocha, C., Zerpa, R., Patiño, L., Valencia, A., Camiña, M., Guevara, J., Lopez, M., Chuquiray, N., Salazar-Lindo, E., Calampa, C., Casapia, M., Meza, R., Bernal, M., Tilley, D., Gregory, M., Maves, R., Hall, E., ... Kasper, M. (2012). Campylobacter antimicrobial resistance in Peru: A ten-year observational study. *BMC Infectious Diseases*, 12(1), 1–7.
- Rahimi, E., & Ameri, M. (2011). Antimicrobial resistance patterns of campylobacter spp. isolated from raw chicken, Turkey, quail, partridge, and ostrich meat in Iran. *Food Control*, 22(8), 1165–1170.
- Reich, F., Valero, A., Schill, F., Bungenstock, L., & Klein, G. (2018). Characterisation of campylobacter contamination in broilers and assessment of microbiological criteria for the pathogen in broiler slaughterhouses. *Food Control*, 87, 60–69.
- Sandberg, M., Sørensen, L. L., Steenberg, B., Chowdhury, S., Ersbøll, A. K., & Alban, L. (2015). Risk factors for campylobacter colonization in Danish broiler flocks, 2010 to 2011. *Poultry Science*, 94(3), 447–453.
- Sattar, S., Hassan, M. M., Islam, S. K. M., Alam, M., Al Faruk, M. S., Chowdhury, S., & Saifuddin, A. K. M. (2014). Antibiotic residues in broiler and layer meat in Chittagong district of Bangladesh. *Veterinary World*, 7(9), 738–743.
- Sakaridis, I., Ellis, R. J., Cawthraw, S. A., Van Vliet, A. H., Stekel, D. J., Penell, J., Chambers, M., La Ragione, R. M., & Cook, A. J. (2018). Investigating the association between the caecal microbiomes of broilers and campylobacter burden. *Frontiers in Microbiology*, 9, 927.
- Sibanda, N., McKenna, A., Richmond, A., Ricke, S. C., Callaway, T., Stratakis, A. C., Gundogdu, O., & Corcionivoschi, N. (2018). A review of the effect of management practices on campylobacter prevalence in poultry farms. *Frontiers in Microbiology*, 9, 2002.
- Skarp, C. P. A., Hänninen, M. L., & Rautelin, H. I. K. (2016). Campylobacteriosis: The role of poultry meat. *Clinical Microbiology and Infection*, 22(2), 103–109.
- Sommer, H. M., Heuer, O. E., Sørensen, A. I. V., & Madsen, M. (2013). Analysis of factors important for the occurrence of campylobacter in Danish broiler flocks. *Preventive Veterinary Medicine*, 111(1–2), 100–111.
- Suzuki, H., & Yamamoto, S. (2009). Campylobacter contamination in retail poultry meats and by-products in the world: A literature survey. *Journal of Veterinary Medical Science*, 71(3), 255–261.
- Taniuchi, M., Sobuz, S. U., Begum, S., Platts-Mills, J. A., Liu, J., Yang, Z., Wang, X. Q., Petri, W. A., Jr., Haque, R., & Houpt, E. R. (2013). Etiology of diarrhea in Bangladeshi infants in the first year of life analyzed using molecular methods. *The Journal of Infectious Diseases*, 208(11), 1794–1802.
- Vanderzant, C., & Splittstoesser, D. (1992). *Compendium of methods for the examination of foods*. American Public Health Association.
- Vinueza-Burgos, C., Wautier, M., Martiny, D., Cisneros, M., Van Damme, I., & De Zutter, L. (2017). Prevalence, antimicrobial resistance and genetic diversity of *Campylobacter coli* and *Campylobacter jejuni* in Ecuadorian broilers at slaughter age. *Poultry Science*, 96(7), 2366–2374.
- Wieczorek, K., Szewczyk, R., & Osek, J. (2012). Prevalence, antimicrobial resistance, and molecular characterization of *Campylobacter jejuni* and *C. coli* isolated from retail raw meat in Poland. *Veterinární Medicina*, 57(6), 293–299.
- Williams, L. K., Fonseca, B. B., & Humphrey, T. J. (2016). *Campylobacter jejuni* in poultry: A commensal or a pathogen? In B. Fonseca, H. Fernandez, & D. Rossi (Eds.), *Campylobacter spp. and Related Organisms in Poultry* (pp. 75–87). Springer.
- Zeng, D., Chen, Z., Jiang, Y., Xue, F., & Li, B. (2016). Advances and challenges in viability detection of foodborne pathogens. *Frontiers in Microbiology*, 7, 1833.
- Zhu, K., Jin, H., He, Z., Zhu, Q., & Wang, B. (2006). A continuous method for the large-scale extraction of plasmid DNA by modified boiling lysis. *Nature Protocols*, 1(6), 3088–3093.

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